Paramdeep Singh Gill

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Education

Thapar Insitute of Engineering and Technology (TIET) — Patiala, Punjab
Bachelor of Engineering in Computer Engineering, CGPA: 8.09 — *Sep 2021 – Jun 2025*

Apeejay School, Mahavir Marg (C.B.S.E. affliated) — Jalandhar, Punjab
10 + 2 (Non Medical), Percentage: 95.60% — *2021*

Apeejay School, Mahavir Marg (C.B.S.E. affliated) — Jalandhar, Punjab
10, Percentage: 90.33% — *2019*

Experience

IIT Ropar — Rupnagar, Punjab
Research intern — *January 2025 – June 2025*
- Explored Knowledge Distillation for Model Compression in Medical Domain. Under the guidance of Dr. Deepti Reddy Bathula (Indian Institute of Technology (IIT) Ropar, India)

AgNext — Sahibzada Ajit Singh (S.A.S.) nagar, Punjab
Android developer intern — *May 2023 – August 2023*
- Flutter: Built android apps that connect to BLE devices.

Awards and Recognition

Runner-Ups - HackForSustainability — 2023
Organised by Saturnalia Fest (largest in North India) for Project Saarthi — *Patiala, Punjab*

Winner - Smart India Hackathon (SIH) (Senior Software Edition) — Aug 2022
Team Leader and Front-End Developer (Problem Statement under DoSEL, Ministry of Education) — *Karnataka*

Awarded Merit Scholarship – II (Tuition fee waive off) — 2022
From Thapar Insitute of Engineering and Technology — *Patiala, Punjab*

Dr Stya Paul Award for Human Values – 2020 — 2020
In Senior Secondary Category and scholarship at Apeejay School, Mahavir Marg — *Jalandhar, Punjab*

ATL Student Innovator Award — 2019
Certificate of Merit by Atal Innovation Mission, Niti Aayog — *Jalandhar, Punjab*

Qualified for National Championship of World Robot Olympiad (WRO) — 2016 and 2017
For the year 2016 & 2017. Secured third position in Regional Championship of WRO 2017.

On-going Projects

XAI-FL — 2026
Research project — *Remote*
- Trying to integrate XAI (Explainable Artificial Intelligence) on models trained using FL (Federated Learning). With a focus on using privacy preserving techniques like Differential Privacy.
- Guide: Dr. Parteek Kumar Bhatia (WSU, USA) and Dr. Sachin Kansal (TIET, India)
- Technology used: PyTorch, InterpretML, NVFlare.

BioFL — 2026
Research project — *Remote*
- Working on applying Federated Learning (FL) on unexplored biomedical datasets.
- Guide: Dr. Parteek Kumar Bhatia (WSU, USA) and Dr. Kashish Goyal (TIET, India)
- Technology used: PyTorch, InterpretML.

Deepfake Detection — 2026
Group Research project — *Remote*
- Working on building a model to detect deepfake in dataset having audio video modality.
- Guide: Dr. Shefali Arora Chouhan (NIT Jalandhar, India)
- Technology used: PyTorch

Projects

ThaparSat research project 2024
Solo project *Patiala, Punjab*

- Systematic Approach to Enhance Signal Quality and Noise Reduction in Remote Sensing and Astrophysical Applications
- Capstone project under the guidance of Dr. Seema Bawa and Dr. Mamta Gulati

Vriddhi App 2022
Group Project *Patiala, Punjab*

- An app to guide parents to facilitate and ensure holistic growth in cognitive, psychomotor and affective domains. Contributed in the front-end of the app (using Flutter-Dart).
- Technology used: Flutter-Dart, Firebase, GetX, React Js, Chart Js, Swiper Js, Figma.

Technical & Research Skills

Machine Learning & AI: PyTorch, InterpretML, NVFlare, OpenCV, NumPy, Matplotlib, Knowledge Distillation, Federated Learning
Programming Languages: C, C++, Python, Dart, HTML, CSS, SQL
Research & Academic Tools: LaTeX, AsciiDoc, Zotero, Git, GitHub, VS Code, JupyterLab
Development Frameworks: Django, Flutter, Firebase

Extra-Curricular and Volunteer

International Conference on Statistical Learning (ICSL) 2025
Presented 'Systematic Approach to Enhance Signal Quality...' *Rupnagar, Punjab*

Open Web Application Security Project (OWASP) 2022 - 2023
Team Member: Presented a session on blockchain as part of orientation of first year students. *Patiala, Punjab*

Center for Training and Development (CTD) 2021 – 2023
Senior Coordinator: Coordinated the Placement Symposium. *Patiala, Punjab*

International Award for Young People (IAYP) 2019
Successfully completed Bronze Category Course. *Jalandhar, Punjab*

References

Dr. Deepti Reddy Bathula bathula@iitrpr.ac.in
Associate Professor, IIT Ropar *Rupnagar, Punjab*

Dr. Seema Bawa seema@thapar.edu
Professor, TIET *Patiala, Punjab*